

Burden and Spacing Influence in Ground Vibration Attenuation at Coal Overburden Blast

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ABSTRACT: Coal mine overburden blasting in Brazil is characterized by a high degree of confinement. Overburden is mainly composed by sedimentary rocks, such as siltstones. These types of rock do not need much energy to be broken and excavated and usually powder factor used is very low, in the order of 150 g/m^3 , resulting in large burden and spacing. Coal blasting is different as coal shots have higher powder factor and tighter burden and spacing to ensure coal fragmentation. Because of that, for the new B3 Mine Copelmi Project, the mine management was concerned about the possibility of higher vibrations close to a large community. Attenuation equations were obtained for siltstone overburden and coal using seismograph monitoring and soon it was realized that coal blasting produced much less vibrations than overburden, resulting in lower attenuation equation slope and confinement coefficient (y-intercept) when compared to overburden blasts. It is assumed that because coal shots are less confined the resultant vibrations are lower as less energy is transferred to the rock mass. After this assumption, changes in the blast design geometry were applied and several tests were conducted in the siltstone overburden and the resultant attenuation equations were compared to the previous equations obtained. This paper presents the blast geometry design changes and the optimum results obtained simply modifying the blast confinement.

1 INTRODUCTION

Like for many mine operations around the world vibration and air-overpressure levels generated during blasting activities are a major issue at the B3 mine, operated by Copelmi Mineração Ltda, located at the town of Butiá, in the southernmost state of Brazil. Blasting operations, in the near future, will be as close as 100 meters from the neighboring residential area and for this reason mine management has decided to start testing different blast geometries and changes in the blast design with the intention of reducing vibration levels as much as possible in order to avoid many complaints from the community.

The coal deposit in the mine is characterized by the presence of many coal seams intercalated by siltstone layers. Mined coal thickness ranges from 0.30 m to 2.5 m with interburdens ranging from a few centimeters to more than 12 m. Overburden thickness averages 40 m. Materials composing the overburden are soil, red and yellow clays and siltstone. The siltstone thickness averages 25 m and only a portion of about 6.5 m need to be drilled and blasted, as shown in Figure 1. The remaining portion of the overburden is mechanically excavated.

Usually the need of drilling and blasting is determine whenever the excavators cannot dig the

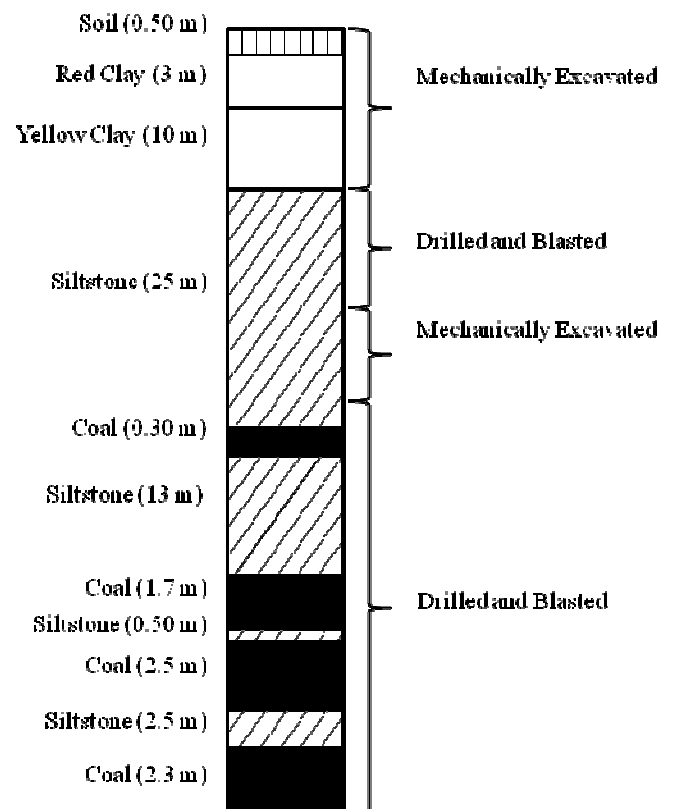


Figure 1. Coal deposit profile.

material too easily, or else, whenever production falls because the material is too hard to be dug there is need of drilling and blasting. Today these layers are predetermined and can vary in thickness as overburden increase.

With the approximation of the mine to the nearby residential area, blasting at the top most siltstone overburden bench will be approximately 100 meters from the structures and vibrations at this distance tend to be high and generate many complaints from the neighbors. With this problem in mind, the mine engineers are trying to determinate the best blast design plan that will reduce vibrations to the minimum possible with the lowest impact in the drilling and blasting operation cost.

2. BLAST DESIGN OVERVIEW

Blastholes are drilled 76 mm in diameter and length varies according to coal or siltstone thickness, for the overburden test bench hole lengths vary from 5.0 to 6.5 m. Burden and spacing are 4.0 by 5.0 m for overburden and interburden siltstone layers and 3.0 by 4.0 m for coal blast. Holes are drilled vertically in a staggered pattern. Explosives used are emulsion cartridges and bulk ANFO. Emulsion cartridges are used as primers to initiate the ANFO column, usually one quarter (15 cm) of a 2.5"x 24" emulsion cartridge is used to initiate the explosive column, Munaretti (2002).

Shock tubes with and without delay periods are used to initiate the explosive columns and on surface connections. Delay periods used have low accuracy, between 23 % and 46 % scatter, making maximum number of holes per delay difficult to predict. More precise delays are available in the Brazilian market but they cost 3 to 4 times more.

Siltstone and coal do not require great amounts of energy to be disaggregated and excavated. For that reason, powder factors used in blast designs are very low, resulting in very confined shots with little rock movement. A minimum amount of energy is used just to induce fractures in the rock mass in order to facilitate the excavation. Another reason for the low powder factor and little movement of the material is to prevent coal dilution which may occur when too much energy is used and there is excessive vertical and horizontal movement. Powder factors range from 70 to 170 g/m³ for overburden and interburdens and 160 to 280 g/m³ for coal. Coal shots use higher powder factor to ensure good coal fragmentation and as the overburden material goes to the waste pile, there is no need to be well fragmented as long as the excavator presents good

productivity during operation and the fragments are no larger than the bucks. This also keeps cost with overburden blasts to a minimum.

Because of the high confinement, vibrations levels generated can be high at close-in distances. Attenuation equations were obtained from data gathered during previous vibration monitoring for coal and overburden blasts and are presented in Figure 2 (Rosenhaim et al. 2012).

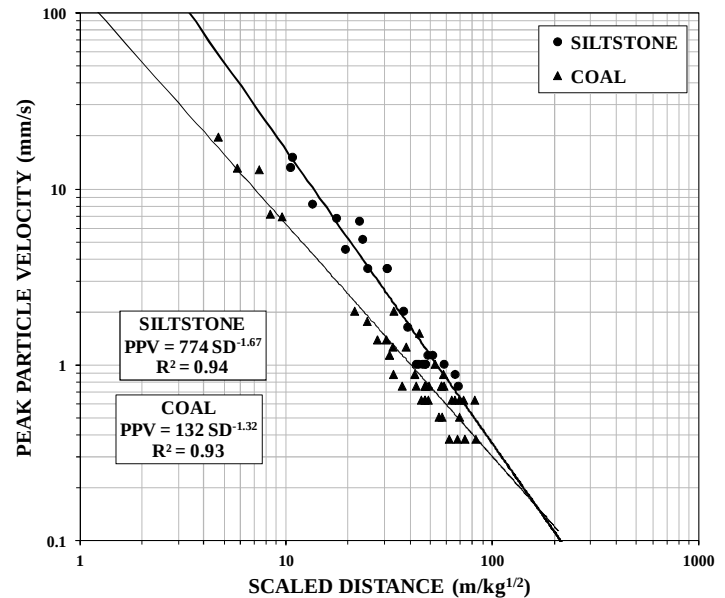


Figure 2. Attenuation equations for overburden and coal blasts.

The attenuation equations represent the best-fit lines through the data and are obtained using a power curve of the form:

$$PPV = a \cdot SD^{-b} \quad (1)$$

Where, “a” is the y-intercept at SD equals to 1 and “b” the slope of the attenuation line. The “a” factor measures the intensity of the seismic energy that is transferred into the ground and is propagated away. It is a function of confinement, explosives density and energy. The slope term “-b” is related to the geology through which the seismic wave passes and is a measure of the decay of the velocity intensity with distance, Downdin (1985).

The equations show that overburden blasts have a higher confinement coefficient (y-intercept) than coal shots, as already mentioned, this is a result of the large burden and spacing, resulting in lower powder factors and high blast confinement. Coal shots also have a lower attenuation due to the depth of the shots, besides attenuation of ground motions with distance there is attenuation with depth as coal shots are located approximately 40 m from the surface.

3. RESULTS

Different drill pattern geometries were tested for overburden blast to determine the best design that would result in good rock breakage with low cost and most important vibration levels within the safe limits set by the Brazilian standards (15 to 20 mm/s from 4 to 15 Hz, 20 to 50 mm/s from 15 to 50 Hz and 50 mm/s above 40 Hz) at close-in distances. For the current blast, vibrations levels are lower than the set limits of 15 mm/s, but mine management wants to reduce vibration way lower than the limits to ensure that complaints will be to a minimum. Peak frequencies are usually between 4 and 15 Hz and are assumed to be result from the influence of the presence of an underground mine in the area. The delay period combination used results in high frequency, in the order of 32 Hz.

In the initial tests, small changes in the drill pattern geometry were done, from a squarer pattern, 4.0 x 5.0 m, to more rectangular or elongated patterns, 3.5 x 5.5 m and 3.5 x 6.0 m. Later on, the same geometry used in the coal blast design was tested, 3.0 x 4.0 m. Even though it was known would result in higher costs, the intention was to evaluate the results regarding vibration levels.

The last test conducted was the introduction of a 150 g booster as initiation charges to the explosive column instead of the quarter part of an emulsion cartridge. Boosters are not commonly used in Brazil to initial small blasthole diameters, mostly because the cost is higher than the emulsion cartridges.

For each change in blast design a series of blasts were conducted and vibration monitoring was carried out. Attenuation equations were obtained from the data gathered are compared in Figure 3.

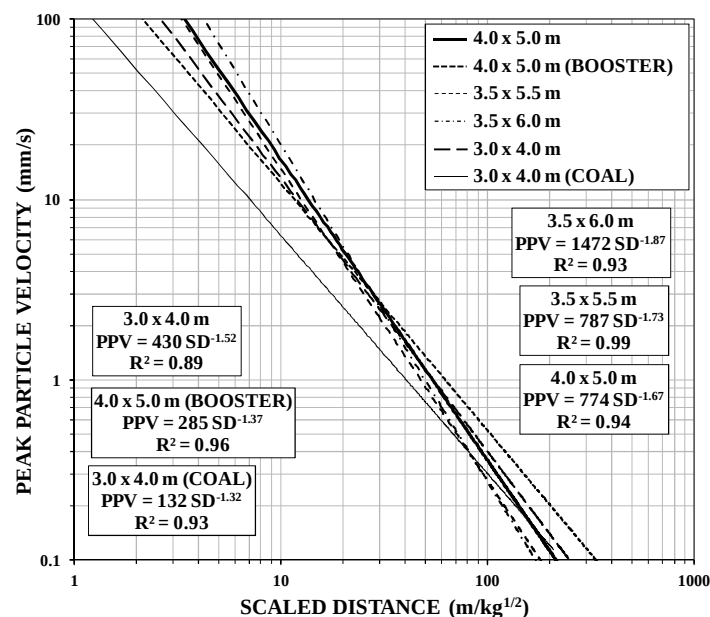


Figure 3. Attenuation plot comparing test results.

As it was expected a decrease in confinement coefficient can be seen as burden and spacing are reduced. When using the same blast design as coal shots, vibration levels produced by the overburden blast were not reduced as much as expected. This is associated to the different rock types and the depth of the blast benches. Even though there is a decrease in the attenuation level the introduction of this blast geometry will result in an increase in cost of 55% to overburden blast costs.

The use of boosters with the current blast geometry presented the best results regarding vibration levels. With the introduction of boosters cost will increase around 12% and the confinement coefficient reduces 63%.

Figure 4 shows the Brazilian standards limits for vibration where data recorded at a stand-of-distance of 100 m for the tests realized are plotted.

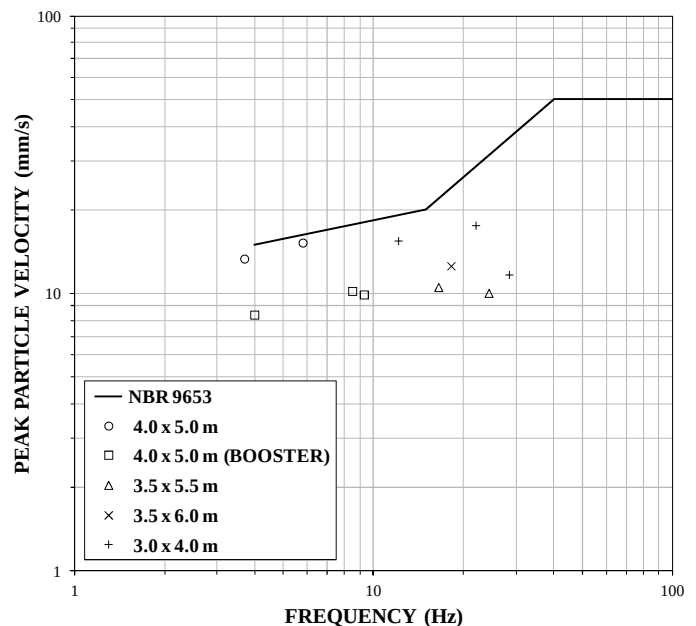


Figure 4. Brazilian standard limits for ground vibrations from blasting, data at stand-of-distance of 100 m.

The graphic in Figure 4 confirms that the best results were obtained when using boosters in the current drill pattern geometry and when burden was reduced.

Table 1 is a summary of average powder factor, average and maximum charge weight per delay and confinement coefficient (a) of the attenuation equations for each blast design tested. Figure 5 confirms that, disregarding the booster test, lower the powder factor is higher will be the blast confinement.

Table 1. Summary of powder factor and confinement coefficient for the blast designs tested.

Blast Design	Powder Factor (kg/m ³)	Charge Weight per delay (kg)		Confinement Coefficient
		Ave.	Max.	
4.0 x 5.0 m (BOOSTER)	0.140	126	81	285
3.0 x 4.0 m	0.247	181	118	430
4.0 x 5.0 m	0.140	213	114	774
3.5 x 5.5 m	0.151	201	124	787
3.5 x 6.0 m	0.133	168	93	1472

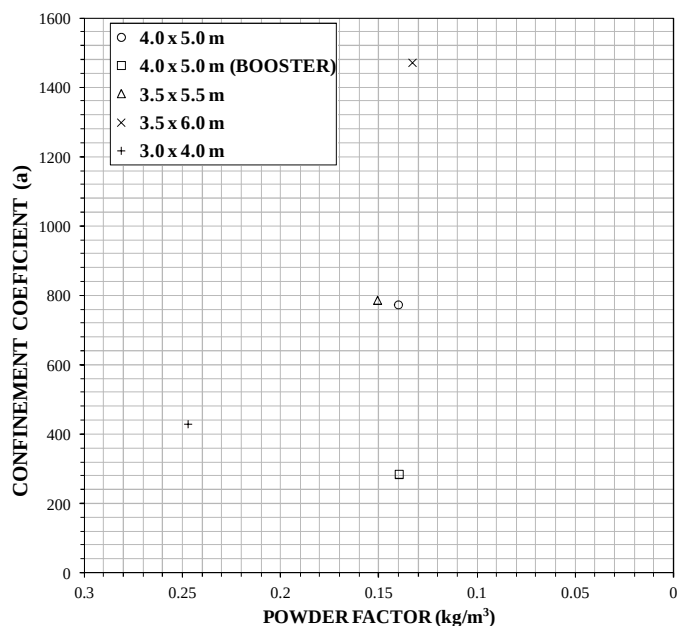


Figure 5. Graphic comparing powder factor and confinement coefficient.

4. CONCLUSIONS

After conducting a series of tests the mine management was able to reduce vibration levels without increasing drilling and blasting costs. The initial drill pattern tested presented small changes in burden and spacing with low cost variations and some improvements in blast vibrations. Applying the same drill pattern used in coal blasts resulted in lower vibrations but with a high cost. Overburden blast represents 40% of the total drilling and blasting operation cost. 55% cost increase in the overburden blast will result in 22% increase the total operation cost.

The use of booster charges presented the best results regarding vibration levels, reducing 63% confinement with low cost increase. The application of the booster charges will be extended to the entire operation as it will result in much better fragmentation in the coal blast as more energy is introduced in the explosives column initiation. Furthermore, costs will increase around 12% for the operation, but it can be reduced as changes to the coal blast design can be applied without impairing fragmentation.

New changes will be applied to blast design with the application of booster to evaluate how much more vibration levels can be lowered and the impact this changes will have in the operation cost.

5. REFERENCES

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